Human Water Cycle

Summer STEM Academy

Oklahoma State Regents of Higher Education



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Human Water Cycle Academy
Student Expectations

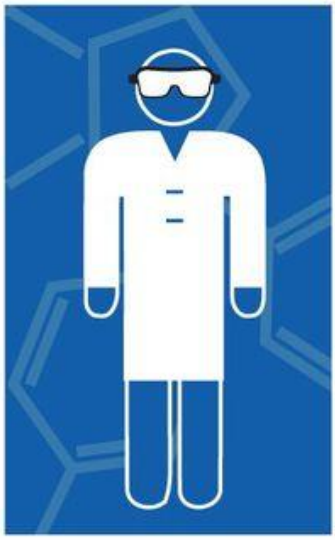
Student Name _____

Academy Site (circle one): ECU Rose State OU

- Respectful
  - Be on time
  - Be where I am supposed to be
  - Pay attention and listen
  - Follow directions
  - Work well with my lab partner(s)
  - Be understanding of others' differences
  - Use kind words
  - Be forgiving
  - Be honest and trustworthy
- Responsible
  - Have all my "stuff" (in provided drawstring bag)
  - Dress appropriately for water/wastewater plant tours
    - Closed-toed shoes
    - Pants/skirt that covers legs
    - (Can bring along and change)
  - Fully read and *follow* directions in LabNotebook
  - Admit to mistakes
  - Clean up after myself
  - Use my *cell phone* only for
    - lab/field work (apps)
    - pictures for Academy
    - video for Academy
    - emergencies

Student Signature _____ Date _____

Parent Signature _____ Date _____



Lab Safety Rules

Science labs offer great opportunities for learning, teaching, and research. They also pose hazards that require proper safety precautions.



Stay safe when conducting your labs by following these guidelines.



Dress appropriately

Tie back long hair, and wear suitable gloves, goggles, and other protective equipment.

Proper supervision

Don't perform lab experiments without instructor supervision (unless given permission to do so).





Know location of emergency numbers & safety equipment

Know the location of safety equipment and emergency phone numbers (such as poison control) so you can access them quickly if necessary.





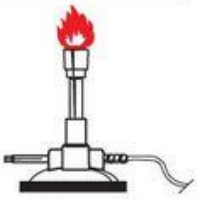
No food

Don't eat or drink in the lab—and never taste chemicals.



ID hazards

Identify hazardous materials before beginning labs.



Be attentive

Be attentive while in the lab. Don't leave lit Bunsen burners unattended or leave an experiment in progress.

Be careful when handling hot glassware

Turn off all heating appliances when not in use. Keep flammable objects away from your workspace.









Keep a clean workspace

Don't obstruct work areas, floors, or exits. Keep coats, bags, and other personal items stored in designated areas away from the lab. Don't block sink drains with debris.



Handle glassware carefully

Properly dispose of anything that breaks. Report cuts, spills, and broken glass to your instructor immediately.



Clean up

After completing the lab, carefully clean your workspace and the equipment, and wash your hands.

CAROLINA®
www.carolina.com

Sources: Carolina Biological Supply Company. "Lab Safety Dos and Don'ts for Students." http://www.carolina.com/teacher-resources/Interactive/lab-safety-instructions/tr11076.tr.

Testing Water

Objectives: Test a potential water source for various water quality parameters and pollutants.

Today you will test water for using different kits:

pH
Turbidity
Dissolved oxygen
Orthophosphate
Nitrate
Alkalinity

With each kit, there are booklets with directions, but there are directions below that you can follow. Record your results in the table on Page

pH

Test procedure—pH (4–10 pH units)



1. Fill two tubes to the first line (5 mL) with sample.

2. Put one tube into the left opening of the color comparator box.

3. Add 6 drops of wide range pH indicator solution to the second tube.

4. Swirl to mix.

5. Put the second tube into the color comparator box.

6. Hold the color comparator box in front of a light source. Turn the color disc to find the color match.

7. Read the result in pH units in the scale window.

Turbidity

6.4.2 Turbidity measurement procedure

Note: *Before a measurement is taken, always make sure that the sample is homogeneous throughout.*



1. Collect a representative sample in a clean container. Fill a sample cell to the line (about 15 mL). Take care to handle the sample cell by the top. Cap the cell.

2. Wipe the cell with a soft, lint-free cloth to remove water spots and fingerprints.

3. Apply a thin film of silicone oil. Wipe with a soft cloth to obtain an even film over the entire surface (Apply silicone oil to a sample cell on page 20).



4. Push the **Power** key to turn the meter on. Place the instrument on a flat, sturdy surface.

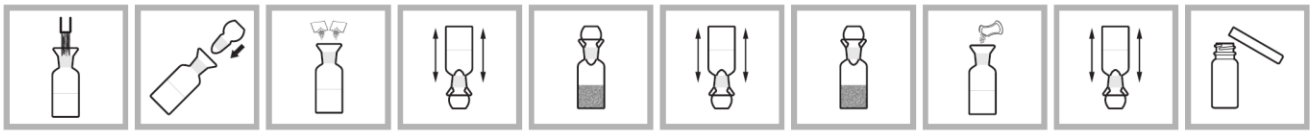
Note: *Do not hold the instrument while making measurements.*

5. Gently invert and then insert the sample cell in the instrument cell compartment so the diamond or orientation mark aligns with the raised orientation mark in front of the cell compartment. Close the lid.

6. Push **Read**. The display shows Stabilizing then the turbidity in NTU (FNU). The result is shown and stored automatically. Additional information is available on the manufacturer's website.

Dissolved oxygen

High range (1 to 20 mg/L) test procedure



1. Fill the BOD bottle (round bottle with glass stopper) with sample water by allowing the sample water to overflow the bottle for 2–3 minutes. Avoid turbulence and bubbles in the sample while filling.

2. Incline the bottle slightly and stopper the bottle carefully to avoid trapping air bubbles. If bubbles become trapped, discard the sample and repeat the test.

3. Remove the stopper and add one Dissolved Oxygen 1 Reagent Powder Pillow and one Dissolved Oxygen 2 Reagent Powder Pillow. Stopper the bottle carefully to avoid trapping air bubbles.

4. Invert the bottle several times until the powders are dissolved. Flocculent (floc) precipitate will form. Brownish-orange precipitate indicates oxygen is present.

5. Wait for floc to settle to approximately half the bottle volume. Floc will slowly settle if high concentrations of chloride are present. In this case, wait 4–5 minutes before proceeding.

6. Invert the bottle again to mix.

7. Wait for floc to settle to approximately half the bottle volume. Floc will slowly settle if high concentrations of chloride are present. In this case, wait 4–5 minutes before proceeding.

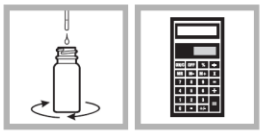
8. Remove the stopper and add one Dissolved Oxygen 3 Reagent Powder Pillow. Stopper the bottle.

9. Invert the bottle several times. Floc will dissolve and the sample will turn yellow if oxygen is present.

10. Add one full measuring tube of sample to the bottle.

Note: *Save the rest of the prepared sample for the low range test, if necessary.*

High range (1 to 20 mg/L) continued



11. Add Sodium Thiosulfate Solution by drops. Count the drops until the color changes from yellow to colorless. Swirl to mix after each drop.

12. The number of drops is equal to the test result in mg/L.

Note: *If the result is 3 mg/L or less, it is advisable to perform a more sensitive test. Follow low range test procedure.*

Low range (0.2 to 4 mg/L) test procedure



1. Use the prepared sample left from Step 9 of the High Range Test. Pour off the contents of the BOD bottle until the level reaches the 30-mL mark on the bottle.

2. Add Sodium Thiosulfate Solution by drops. Count the drops until the color changes from yellow to colorless. Swirl to mix after each drop.

3. The number of drops multiplied by 0.2 is equal to the test result in mg/L.

Alkalinity



1. Fill the measuring tube with sample.

2. Pour the sample into the mixing bottle.

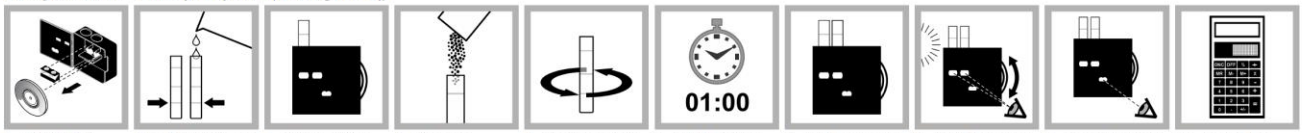
6. Add one Bromcresol Green-Methyl Red Powder Pillow. Swirl to mix.

7. Add the 0.035 N Sulfuric Acid Standard Solution by drops. Mix after each drop. Count the drops until the color changes from green to pink.

9. Multiply the total number of drops by 20 to get the total (methyl orange) alkalinity result as $CaCO_3$.

Phosphate

Test procedure—Orthophosphate (0–4 mg/L PO_4)



1. If installed, remove the long-path adapter.

2. Fill two tubes to the first line (5 mL) with sample.

3. Put one tube into the left opening of the color comparator box.

4. Add one PhosVer 3 Phosphate Reagent Powder Pillow to the second tube.

5. Swirl to mix. A blue color develops.

6. Wait 1 minute. Read the result within 5 minutes.

7. Put the second tube into the color comparator box.

8. Hold the color comparator box in front of a light source. Turn the color disc to find the color match.

9. Read the value in the scale window.

10. Divide the value by 10 to get the result in mg/L.

	Location A	Location B	Location C
Visual description of the water			
pH			
Turbidity (NTU)			
Alkalinity (m/L @ $CaCO_3$)			
Dissolved Oxygen (mg/L)			
Phosphate (mg/L)			

Grow Cultures from Your Water Sample

Objective: Grow cultures from your water sample for further microscope observation.

1. If a Petrifilm plate pack has been stored in the refrigerator, let the package come to room temperature before opening it. This step prevents condensation from forming inside the package.
2. Place the Petrifilm plate on a level surface, with the grid side down. Lift the top film (fig. 2).

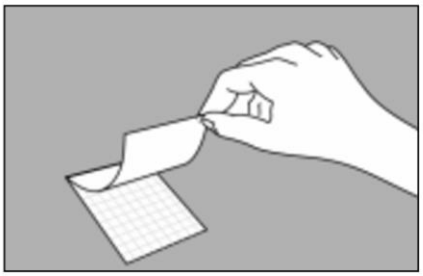


Figure 2

3. With pipet perpendicular to the Petrifilm plate, place 1 mL of sample onto the center of the bottom film (fig. 3). If necessary, samples can be diluted with distilled water, liquid culture medium, or buffers with pH between 6.6 and 7.2.* If antibiotics are to be added to the medium, add them to the inoculating liquid at the working concentration.

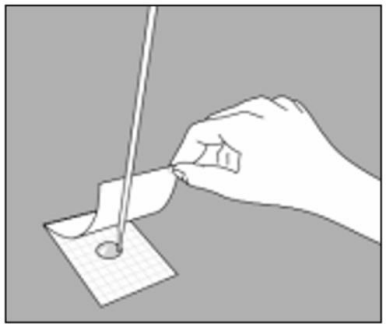


Figure 3

4. Release the top film. Allow it to **drop** onto the bottom film. Do not roll the top film down (fig. 4).

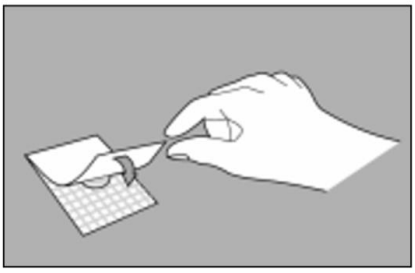


Figure 4

5. Hold the spreader with the circular ridge down (flat side up). Place the spreader on the top film over the inoculum (fig. 5).

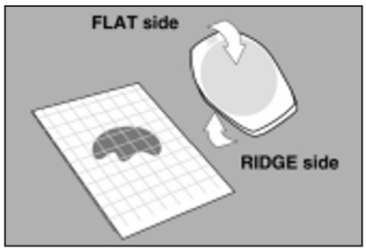


Figure 5

6. **Gently** apply pressure on the spreader to distribute the inoculum over a circular area. Do not twist or slide the spreader (fig. 6).

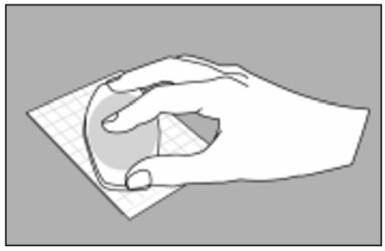


Figure 6

7. Lift the spreader. Wait at least one minute for the gel to solidify (fig. 7).

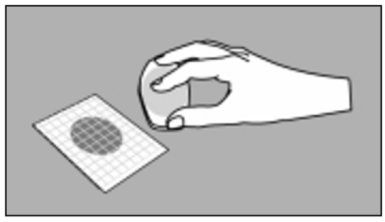


Figure 7

8. Incubate plates with the clear side up in stacks of up to 20 plates. Incubation time and temperature will vary according to the application and equipment available.
9. Colonies on Petrifilm plates can be counted on a standard colony counter or other light source (fig.8). Bacterial colonies on Petrifilm AC plates are red because of the indicator dye in the medium.

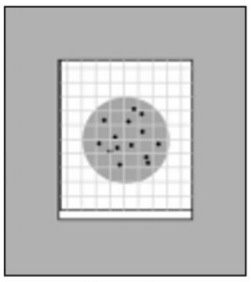


Figure 8



Figure 9

10. Colonies may be isolated for further study or to inoculate additional cultures. Lift the top film and pick the colony from the gel. The medium will adhere to the top film.

Global Positioning System (GPS) and Geographical Information System (GIS)

Objectives
To understand how a GPS works and how it can be combined with GIS to collect features on the landscape to add information to a map.

Materials
Cell phone
ESRI Field Maps app
Computer Lab
ArcGIS Online (AGOL) login with password

Procedure
1. Overview of GPS
2. Overview of GIS
3. Explore the ESRI Field Maps app and definition list
4. Students use GPS to collect campus features as data points with their phones using the ESRI Field Maps app
5. Students analyze the point data in AGOL in the computer lab
6. Overview of mapping in AGOL
7. Students create their own map using data points collected earlier
8. Explanation of how students can create their own ArcGIS Online account

The Unmanned Aerial Vehicle (UAV) aka Drone

Objectives
To understand the uses of an UAV as related to GPS and GIS, learn the basics of flying, and some basic rules for drone use.

Materials
UAV

Procedure
1. Overview of how drones are used with GPS and GIS in the industry
2. Overview of flight basics with drones
3. Overview of rules for drone use (e.g., airspace/property rights, weather)
4. Overview of flight as a team (e.g., responsibilities, communication)
5. Students learn to fly drones with basic maneuvers (e.g., forward, backward, side to side, up, down)
6. Students fly obstacle course using flight rubric for scoring (following page)

Flight Rubric

Teammate 1

Knowledge/Skills	Points
Team communication	/10
Launch	/1
Flight – land on target (points awarded based on rings on target)	/5
Flight – fly through hoop	/3
Flight – successfully fly three obstacles in a row	/10
Controlled landing	/4
Stuck in net (each time)	/-1
Hit obstacle (each time)	/-1

Teammate 2

Knowledge/Skills	Points
Team communication	/10
Launch	/1
Flight – land on target (points awarded based on rings on target)	/5
Flight – fly through hoop	/3
Flight – successfully fly three obstacles in a row	/10
Controlled landing	/4
Stuck in net (each time)	/-1
Hit obstacle (each time)	/-1

Water Treatment Plant

Questions for water treatment trip:

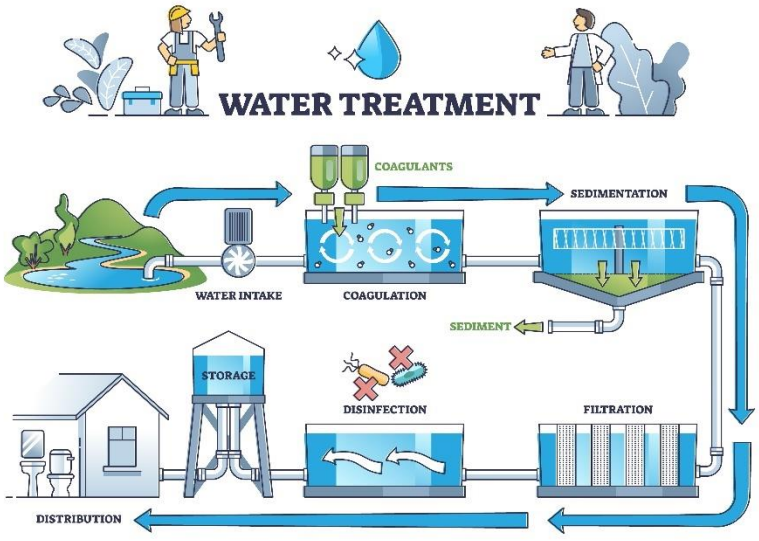
1. Where does this water come from?

2. How much water gets treated here?

3. Do you have to remove anything from the water?
   - Why, Why not?

4. How do you keep pathogens out of the water?

5. How many people drink this water?



Coagulation and Flocculation: Hydrolyzing Metal Salts

Objectives

To conduct jar tests on a natural surface water in order to estimate an optimum dosage of aluminum sulfate for the removal of suspended matter or color. To observe the rate of floc formation and sedimentation.

Apparatus

1. Six-place laboratory stirrer (jar test apparatus)
2. Turbidimeter
3. pH Meter
4. Assorted measurement pipettes (1, 5, 10 ml) and volumetric dispenser to 50 ml
5. Plastic cups for coagulant additions, 10 ml.

Materials

1. Aluminum sulfate solution, 10 mg $Al_2(SO_4)_3 \cdot 14H_2O$/ml, 1 liter
2. Sodium carbonate[1] 10 g/L as $CaCO_{3\ eq}$.
3. Bentonite suspension, 1000 mg/l, made from deionized, 10.8 L.
4. Standard buffer solutions.

Procedure

Part 1: Determination of Optimum Coagulant Dosage

1. Analyze Bentonite Suspension for pH and turbidity. Record both the water temperature at time of test and ambient air temperature.
    - Turbidity: Fill the sample cell to the line with approximately 15-mL of sample. Cap the sample cell. Clean the outside of the cell with a soft, lint-free cloth to remove fingerprints and water spots (if needed apply a thin film of silicone oil and wipe with a soft, lint-free cloth to obtain an even film over the entire surface.) Press I/O button. Put the sample cell in the instrument cell compartment so the white diamond or assigned orientation mark aligns with the raised mark in front of the cell compartment. Close the cover. Select manual or automatic range by pressing the RANGE key. The display will show "AUTO RNG" when the instrument is in the automatic range selection. Select signal averaging mode by pressing the SIGNAL AVERAGE key. The display will show "SIG AVG" when the instrument is using averaging. Use signal average mode if the sample causes a noisy signal (display changes constantly). Press READ. The display will show "- - - - NTU" then the turbidity value in NTU. Record the turbidity after the lamp symbol in the lower left corner of the screen turns off.
    - pH: immerse probe and temperature sensor in solution. Press read on pH meter. Wait for reading to stabilize and record the pH.

2. Place 600 milliliter aliquots in 1-liter beakers on the six-jar laboratory stirrer and check stirrer operation (200 rpm). A light table will facilitate viewing of the contents of the beakers.

3. The initial alkalinity of the water is 0 mg/L as $CaCO_3$ equivalent. Augment the alkalinity of the water by adding 15 mL of Na_2CO_3 using a graduated cylinder to transfer the amount of Na_2CO_3 to each 1 L beaker with sample. Measure and record the pH of each Jar Tester in **Table 1**.

4. Prepare the coagulant dosage using the 50 mL conical containers and 50 mL graduated cylinder. In five different conical containers, measure out 4, 8, 12, 16 and 20 mL of aluminum or ferric sulfate solution. **You will add (in the next step) the coagulant all at once and after alkalinity adjustment.**

5. After alkalinity has been added and the speed has been increased to 200 rpm, now *(quickly)* add 4 to 25 mL of aluminum or ferric sulfate solution to the five beakers keeping one beaker as a control and start a 15 seconds rapid mix at 200 rpm. This step should be done as quickly as possible. Be sure you are ready to add the coagulant and start rapid mix.

6. Flocculate at 60 rpm speed for 10 minutes to represent design conditions. Record the elapsed time before a visible floc is formed. If large flocs are formed, it may be desirable to reduce the paddle speed. Note the size and appearance of the floc formed.

7. After flocculation, remove the paddles and settle for 10 minutes.

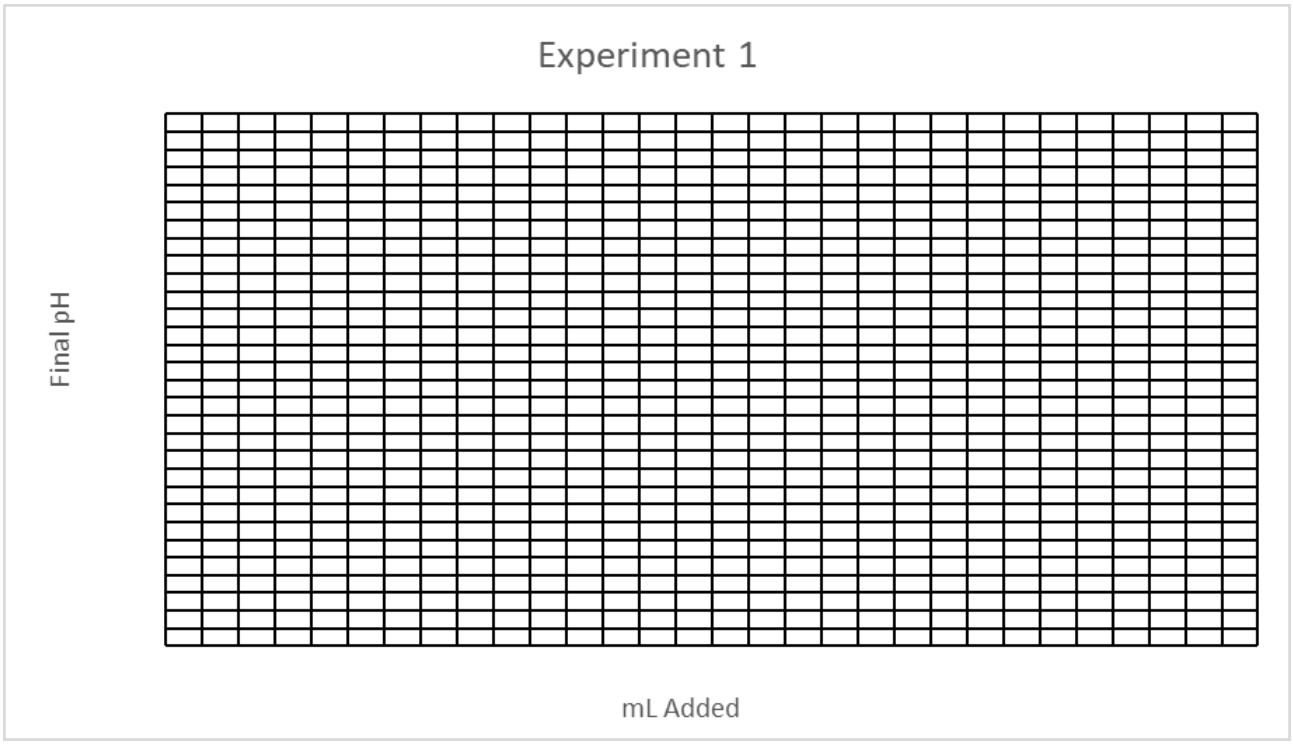
8. Measure the turbidity of the liquid in each jar by sampling at the supernatant (at 2¼" or 6 cm below the water surface), taking care not to disturb the sediment in sampling. Measure the pH of the supernatant and the depth of sludge (use ruler supplied). Estimate the volume of sludge which might be produced by treating 1 million gallons of the test water.

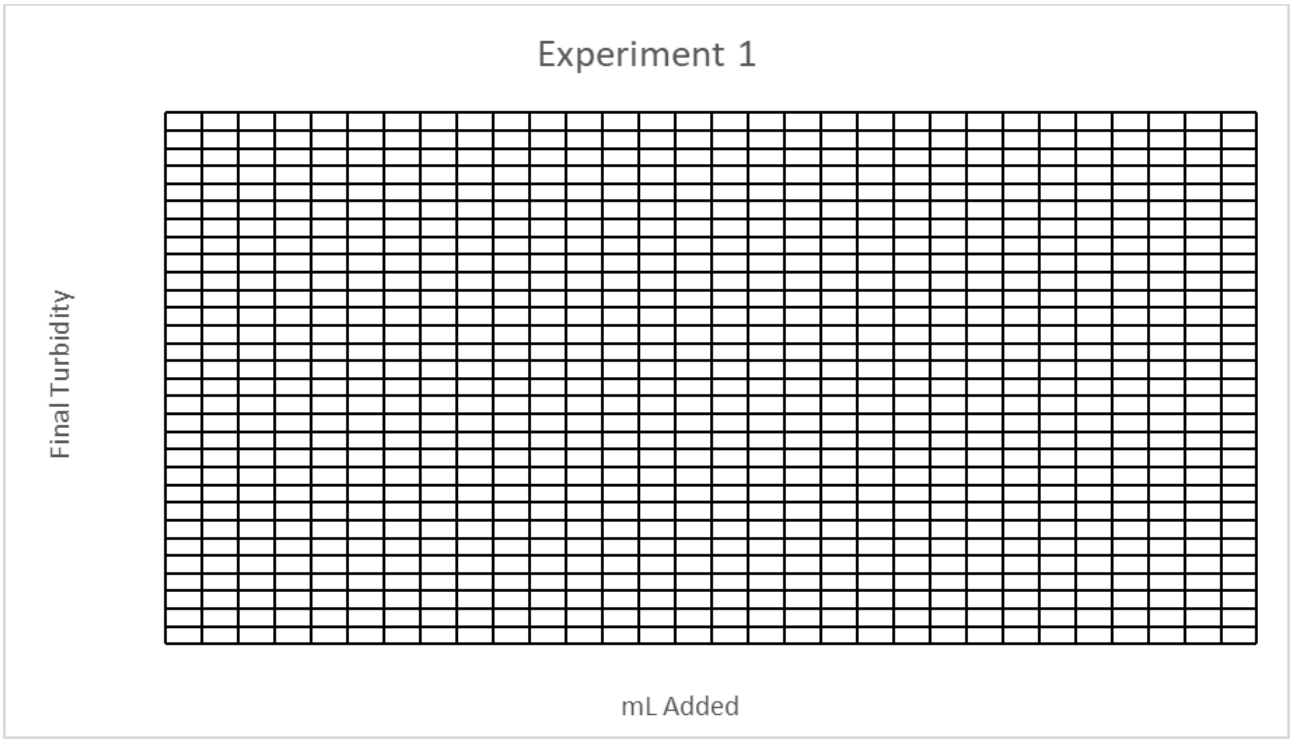
9. Based on these results, select the optimum dosage based on supernatant clarity (both low turbidity) and settleability of floc.

10. If satisfactory results are not obtained, repeat the jar test experiment with another range of coagulant doses. The test also may be repeated, if time allows, to further narrow the range of dosages near the optimum. Estimate the range of dosages over which this coagulant would be effective.

Experiment Table 1

Jars	Initial pH	Final pH	Initial Turbidity (NTU)	Final Turbidity (NTU)
1				
2				
3				
4				
5				
6				


Experiment 1
Final pH
mL Added


Experiment 1
Final Turbidity
mL Added

Part 2: Effects of Settling Time – Type II flocculated particle settling.

1. Type II (or flocculent) settling is sedimentation of suspensions that flocculated particles. All six beakers will have the same chemical additives.

2. No pH adjustment is needed for this experiment. All beakers should be amended with 15 mL of alkalinity before the addition of the optimum coagulant dose. **Analyze Bentonite Suspension in each jar for turbidity, pH and temperature. Record both the water temperature and pH at time of test in Table 2.**

3. In five different containers, measure out the optimum dosage (determined from previous Laboratory 1) of aluminum or ferric sulfate solution. Prepare identical (optimum) coagulant dosages for 4 of the beakers leaving the control blank. Now add the coagulant to each of the four (4) beakers and start a one-minute rapid mix at 200 rpm for ALL FIVE (5) beakers. This step should be done as quickly as possible. Be sure you are ready to add the coagulant and start rapid mix.

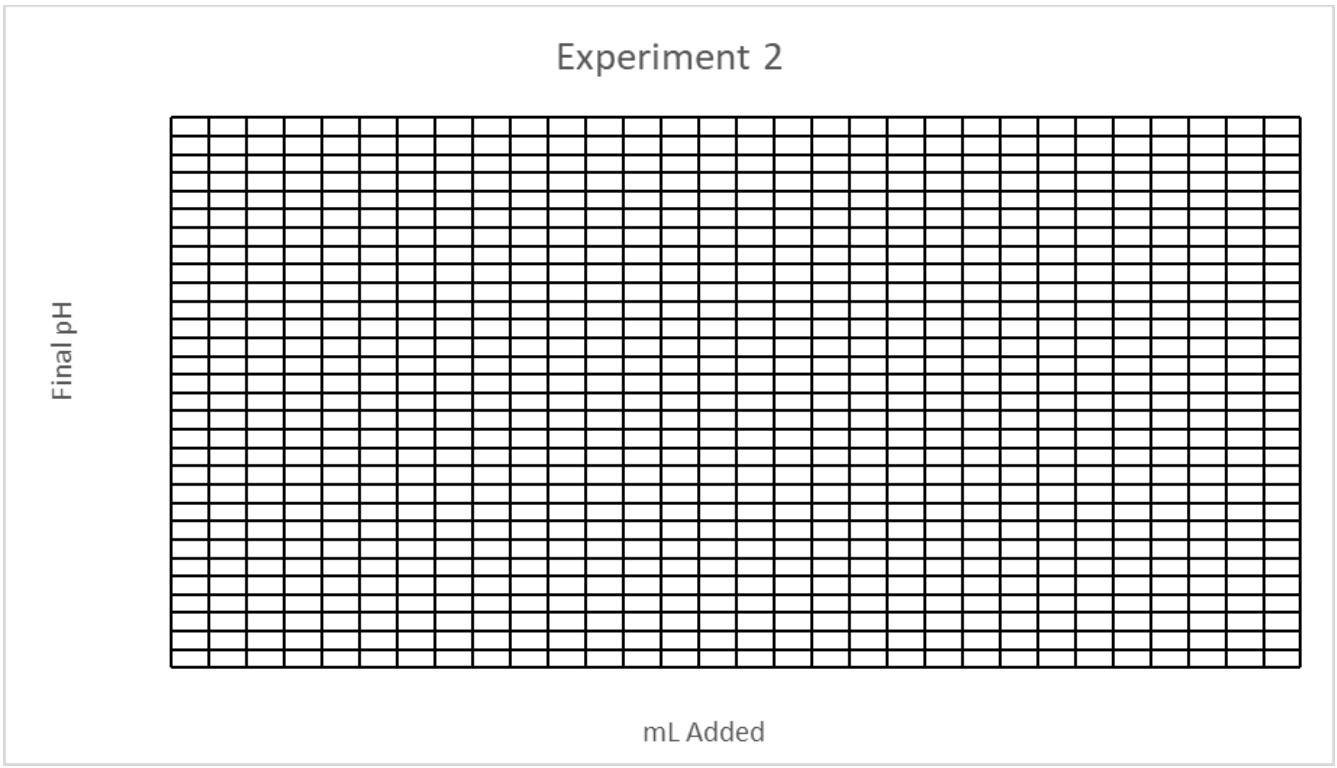
4. After one-minute rapid mix, reduce speed to 40 rpm. Allow flocculation for 10 minutes. At the end of the flocculation time, stop stirring and begin settling cycle. In order to better approximate the clarification, it is good practice to allow a lag time of about 30 seconds from the time of stopping the flocculation to the beginning of settling. This is about the time required for the fast rotary motion to subside. In a plant, it is about the time required for the flocculated water to go from the flocculation to settling basins before settling begins.

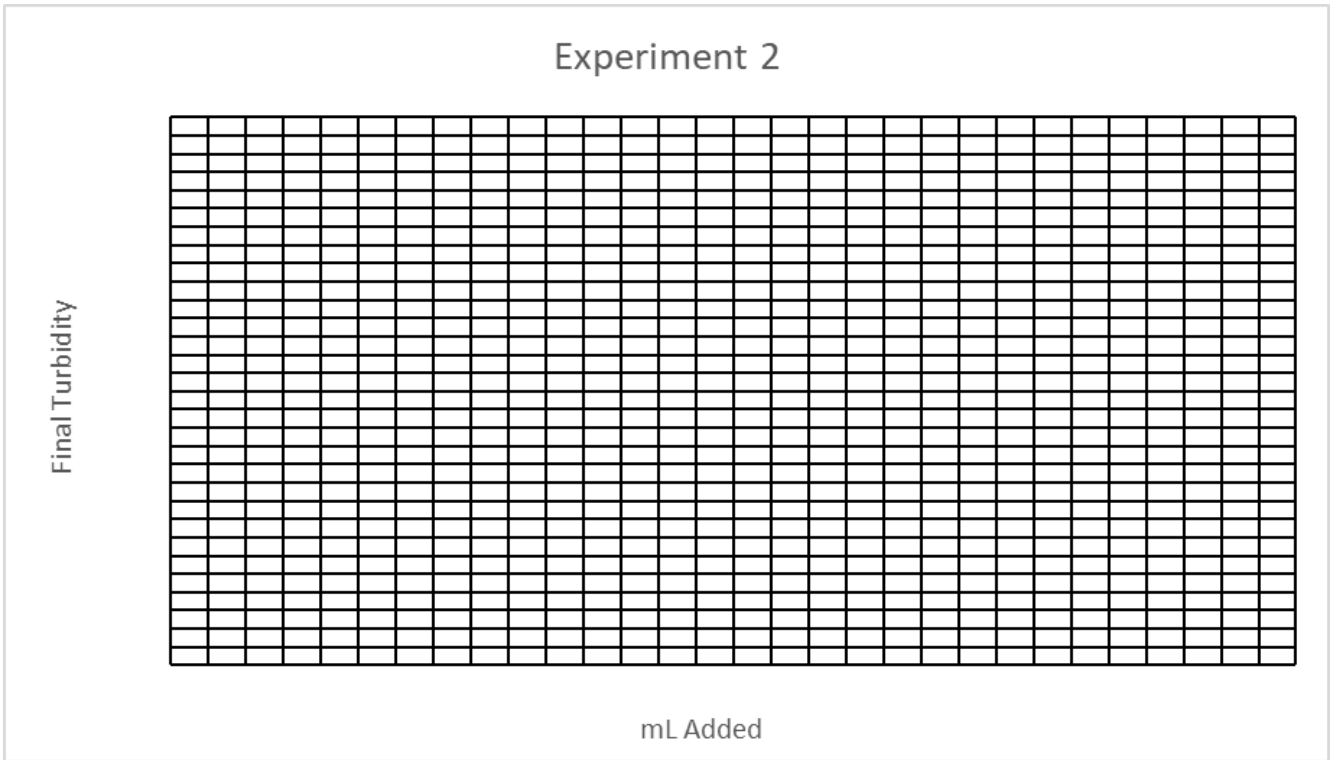
5. **Allow for 4, 8, 12, 16 minutes for settling for the 4 beakers with coagulant. The control will also settle for 16 minutes.** For each settling time, measure turbidity **(LEAVING THE CONTROL LAST)** by pulling a sample from 2¼" or 6 cm below the water surface. Repeat until the end of the total settling time.

6. Measure the depth of sludge in the beaker.

Experiment Table 2

Jars	Initial pH	Final pH	Initial Turbidity (NTU)	Final Turbidity (NTU)
1				
2				
3				
4				
5				
6				


Experiment 2
Final pH
mL Added


Experiment 2
Final Turbidity
mL Added

Computer Project 1: Drinking Water Disinfection

Background

The design of a drinking water disinfection process requires a careful balance between adding enough disinfectant to control microorganisms, without overdosing and forming unacceptably high concentrations of disinfection by-products (DBPs). This balance is affected by many factors, including disinfectant type, contact time available, temperature, pH, and dissolved organic carbon concentration. It is important that drinking water engineers and professionals understand some of the trade-offs between microbial control and DBP formation, and how to approach a treatment design to allow both objectives to be simultaneously met.

This project uses software called WatPRO that simulates the disinfection process. In particular, it calculates the *Giardia* and virus removal and inactivation credits that are achieved by a user-defined treatment train. The control of these organisms is regulated in the United States for facilities treating surface waters. WatPRO further predicts the formation of trihalomethanes, haloacetic acids, and chlorite, DBPs whose concentrations are limited by regulations due to suspected toxicity. It is the engineer's job to design a treatment train that provides adequate *Giardia* and virus control without exceeding DBP limits.

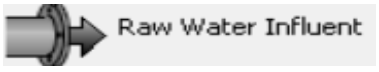
Objective

You will use WatPRO to optimize the disinfection process at a treatment plant. The treatment objectives are to meet *Giardia* and virus reduction targets without exceeding disinfection by-product (DBP) limits (see the "Procedures"). You will write a report justifying your proposed improvements to the treatment train.

Procedures

STEP 1. Input water treatment units. The default treatment train is shown in Figure 1.

Using the left menu, select the **Raw Water Influent** icon (right) and drag to the layout space (below). If the icon is not visible, select the expansion arrow on the left to show all available unit options, then select the ***Influent*** option. This will expand the options under this menu. Look for



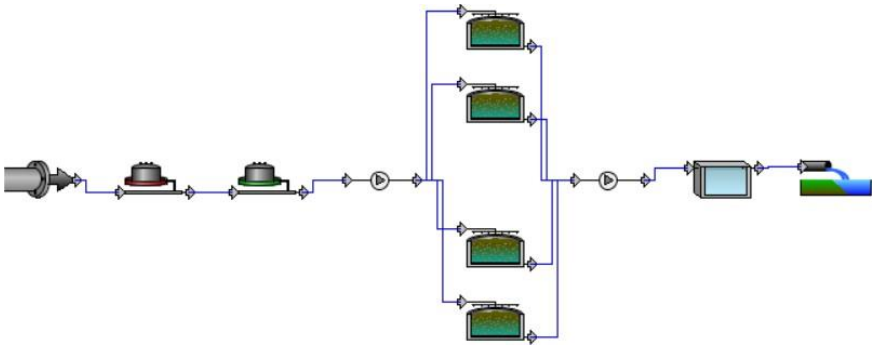
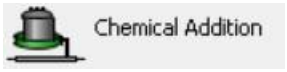


Figure 1. Water treatment plant for direct filtration.

and hover the mouse cursor over this icon. Press and hold the left cursor to drag the icon to the layout space on the right.

Using the left menu, select to expand icons in the ***Chemical*** menu option, now select the **Chemical Addition** icon and drag to the layout space. Chemicals are added to the treatment train to enhance particle removal or adjust water quality parameters. The **Chemical Addition** icon should look similar to this



Using the left menu, select to expand icons in the ***Chemical*** menu option, now select the **Disinfectant Addition** icon and drag to the layout space. Disinfectants are added to the treatment train to inactivate pathogens. The **Disinfectant Addition** icon looks similar to the **Chemical Addition** icon except for the labeling.

Using the left menu, select to expand icons in the ***Distribution*** menu option, now select the **Pass Through Point** icon and drag to the layout space. This unit allows for the different flow streams to connect. The **Pass Through Point** icon should look similar to this



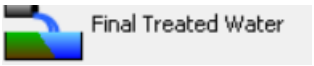
Using the left menu, select to expand icons in the ***Solid Separation*** menu option, now select the **Filtration** icon and drag to the layout space. Be sure to add them in a column similar to figure 1. Repeat this step three additional times. This unit removes small settleable particles in drinking water. The **Filtration** icon should look similar to this

Using the left menu, select to expand icons in the ***Distribution*** menu option, now select the **Pass Through Point** icon and drag to the layout space. Next, select to expand icons in the ***Storage*** menu option, now select the **Clear Well** icon and drag to the layout space. This unit stores treatment water ready for distribution. The **Clear Well** icon should look like this



Using the left menu, select to expand icons in the ***Effluent*** menu option, now select the **Final Treated Water**

icon and drag to the layout space. The **Final Treated Water** icon should look like this



STEP 2. Connect treatment units.

Move the cursor to the connection point of the **Raw Water Influent** icon. The cursor will change from the standard white arrow ( ) to a gray directional arrow ( ). Press and hold the left-mouse clicker and drag to the inlet of connection of the **Chemical Addition** icon. The arrows will light up green when it has been connected properly. This is when you release the mouse. This connects the two units with a pipe. Repeat this process for the remaining units in the treatment sequence. For the **Pass Through Points**, connect each to the inlet of the **Filtration** icons.

Table 1. Data Entry Table for Influent

Influent Water Quality Parameter	
Flow Rate	5000 m^3/d
Temperature	12.0 deg C
pH	6.8
Total Organic Carbon	5.0 mg/L
Turbidity	20.0 NTU
Calcium Hardness (as $CaCO_3$)	80 mg/L
Total Hardness (as $CaCO_3$)	100.0 mg/L
Alkalinity (as $CaCO_3$)	100.0 mg/L
Giardia	2.0 cysts/100L

STEP 3. Set up the Site Information.

Move the cursor to the **Raw Water Influent** icon. While hovering over the icon, use the right-click button on the mouse to **Edit Parameters**. Match the influent water quality parameters to only those found in Table 1. Select *Accept* to save the changes.

Move the cursor to the **Chemical Addition** icon. While hovering over the icon, use the right-click button on the mouse to **Edit Parameters**. From the drop-down menu, select Alum (Aluminum Sulfate) as the *Chemical Type* and 50.0 mg/L as the

Chemical Dosage. Select *Accept* to save the changes.

Move the cursor to the **Disinfectant Addition** icon. While hovering over the icon, use the right-click button on the mouse to **Edit Parameters**. From the drop-down menu, select NaOCl – As CL2 (Anhydrous Sodium Hypochlorite) as the *Disinfectant Type* and 1.5 mg/L as the *Disinfectant Dosage*. Select *Accept* to save the changes.

Move the cursor to ONE of the **Filtration** icons. While hovering over the icon, use the right-click button on the mouse to **Edit Parameters**. From the drop-down menu, select Direct as the *Filter Type*, 50.0 m^3 as the *Volume*, 30.0% as the *TOC Removal Efficiency* and 30.0% as the *UV254 Removal Efficiency*. Select *Accept* to save the changes. Repeat this for each of the four filters.

Finally, move the cursor to the **Clear Well** icon. While hovering over the icon, use the right-click button on the mouse to **Edit Parameters**. Enter 300.0 m^3 as the volume. From the drop-down menu, select Unbaffled as the *Baffling Description* and ensure that the Include in Ct(Sum) box is checked.

STEP 4. Set up the Site Information.

You may have to specify a 2nd order chlorine decay model in some versions of WatPRO. Go to the MODEL menu (located in the upper left corner of the screen), select MODEL CALIBRATION, then CHLORINE DECAY. Click on the DECAY tab, and then specify in the ORDER window, "2_{nd}".

Table 2. Treatment criteria

Water Quality Parameter	
Required Giardia *reduction*	10-logs
Required virus reduction	20-logs
Required Crypto reduction	4-logs
Total THM limit	30 µg/L
HAA5 limit	20 µg/L
ClO2 - limit	1 mg/L
Maximum Residual Cl2 or chloramines in effluent	2 mg/L
Maximum Residual ClO2 in effluent	0.7 mg/L
*Maximum effluent Turbidi*ty	0.5 NTU

STEP 5. Set up the Treatment Criteria.

The water quality needed for drinking water will depend on both the inactivation for pathogens and maximum levels of disinfectant (Table 2). Using the top menu, select the **Model** to pull down the options. Next select **Effluent Limits**.

In the *Data Entry* dialog box, enter the treatment criteria from Table 2. If an entry field is greyed-out, select the checkbox to the left of the target criterion. This will allow data entry. Confirm the appropriate unit of measure is applied. Unit of measures can be modified using the dropdown menu to the right of the criterion. Select *Accept* to save the changes.

STEP 6. Run the Model.

First complete a Connections check. Using the top menu, select the **Model** to pull down the options. Next select **Check Connections**. A pop-up window will indicate that the model is ready for analysis. Next select the **Model** to pull down the options. Now select **Calculate**. Using the **Final Effluent Summary** table at the bottom left, scroll down to check for compliance. Compliance of water quality constituents will be color-coded *green* while non-compliance will be color- coded *red*.

You can obtain more specific information by selecting various categories using the pull-down option in the **Process Output Summary** window. The options include disinfectant dose, Contact-Time Disinfectant Parameter, Pathogen, etc.

STEP 7. Conduct a Chlorine Sensitivity Analysis.

Using the top menu, select the **Analysis** to pull down the options. Next select **Sensitivity Analysis**. A new dialog box will appear. Under **Parameter Selection**, *location, parameter and effect* drop-down menus can be used to narrow the analysis parameters. For **Location** select *Disinfectant Addition*. For **Parmeter** select *Disinfectant Dosage*. This will be the variable along the graph's abscissa. For **View Effect** select *Water*. This will be the variable along the graph's ordinate.

Next, select **Range Settings**. While this is a user input variable, commonly low values are around 0.5 mg/L and high values are around 100 mg/L. The **Number of steps** is related to the mode's iterations. The higher the number (~ 500), the more iterations the model will run with increasing processing time. The lower the number (~10), the quicker the processing time, but the output will become more erratic. For this similar model, 100 iterations will be sufficient.

Then, select **Run Analysis.** A graph will appear showing the effectiveness of your treatment system at removing viruses, giardia, and cryptosporidium. For inactivation, it is typical for the colored

lines to be higher than the dotted horizontal line which indicates the initial treatment criteria inputted in step 5.

Now, select **DBPs-graph** tab located on the left under **Run Analysis**. This graph shows your DBP formation. The goal for this graph is to keep the colored lines underneath the dotted horizontal lines which was our initial treatment criteria inputted in step 5.

STEP 8. Problem Analysis.

Assume that you are a consultant hired by a municipality that is concerned with ensuring adequate control of *Giardia* and viruses, without exceeding DBP limits. The municipality informs you that they would consider making the following adjustments to their present treatment train:

□ Disinfectant dose and point(s) of application

□ The use of chloramines or chlorine dioxide instead of, or in addition to, chlorine

□ The use of a contact chamber (200 m^3 maximum)

□ Building coagulation/flocculation/sedimentation units or other units

□ Increasing the alum dosage up to 50 mg/L

Compare different alternatives and suggest the optimum treatment train to meet the municipality's concerns about microbial control and DBP concentrations given the constraints with which you are presented. Submit your findings in the format of a brief presentation. Remember to use the computer model in a realistic manner, interpreting the output using your professional judgment. If the results look wrong, explain why. Don't assume the model is necessarily correct; it's only a prediction model based on inputs.

Wastewater Treatment

Questions for wastewater treatment trip:

1. How much water does the plant treat?

<u>Preliminary/headwork (smells awful):</u>

2. Why does the material from here go to the landfill?

   2a) Can it all go to the landfill?

<u>Primary Treatment:</u>

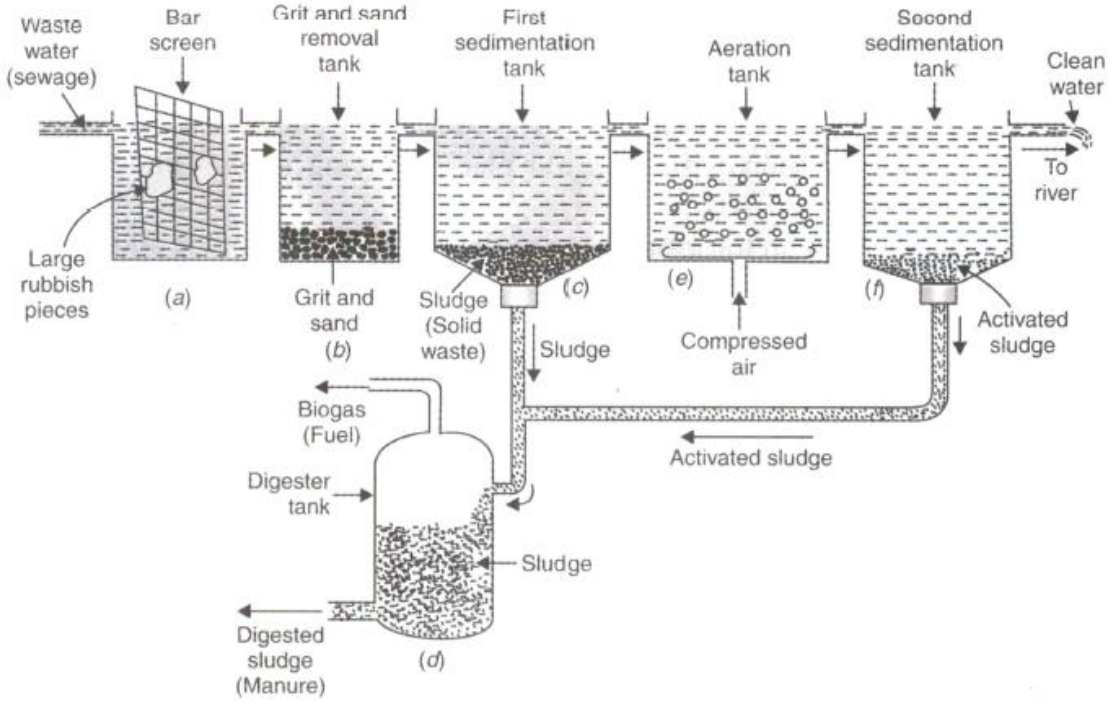
3. What happens in the primary clarifiers?

<u>Secondary Treatment:</u>

4. What eats/ breaks down the poop?

<u>Tertiary Treatment: Disinfection</u>

5. Where does your water go to once treated?



Monitoring Bacterial Growth

Objective: We will grow two cultures in a glucose solution and monitor the growth through glucose consumption.

Supplies:
4 flasks
Glucose solution
Pipettes
Glucose meter
Glucose strips

1) Label flasks “0”, “0.5”, “1.0”, and “1.5”. Put your group initials on each flask with the date.
2) Put 200 ml of the glucose solution in each flask.
3) In the flask labeled “0.5” put 0.5 grams of the yeast. Swirl the flask slowly, not splashing any of the solution.
4) In the flasks labeled “1.0” place 1.0 grams and “1.5” place 1.5 grams of yeast in each. Swirl flask slowly, not splashing any of the solution.
5) In the flask labeled “0”, do not place anything in it. This is your Control. Swirl the flask slowly, not splashing any of the solution.
6) For the initial glucose reading, obtain a dropper of the yeast/glucose solution and place a drop in a weigh boat.
    a. BEFORE YOU TAKE A DROPPER OF THE SOLUTION OUT, BE SURE TO SWIRL YOUR SAMPLE. You want it mixed well.
7) Place a glucose strip in the glucose meter as seen in the figure:



8) The end of the strip will absorb the drop of solution if you touch it to the drop.



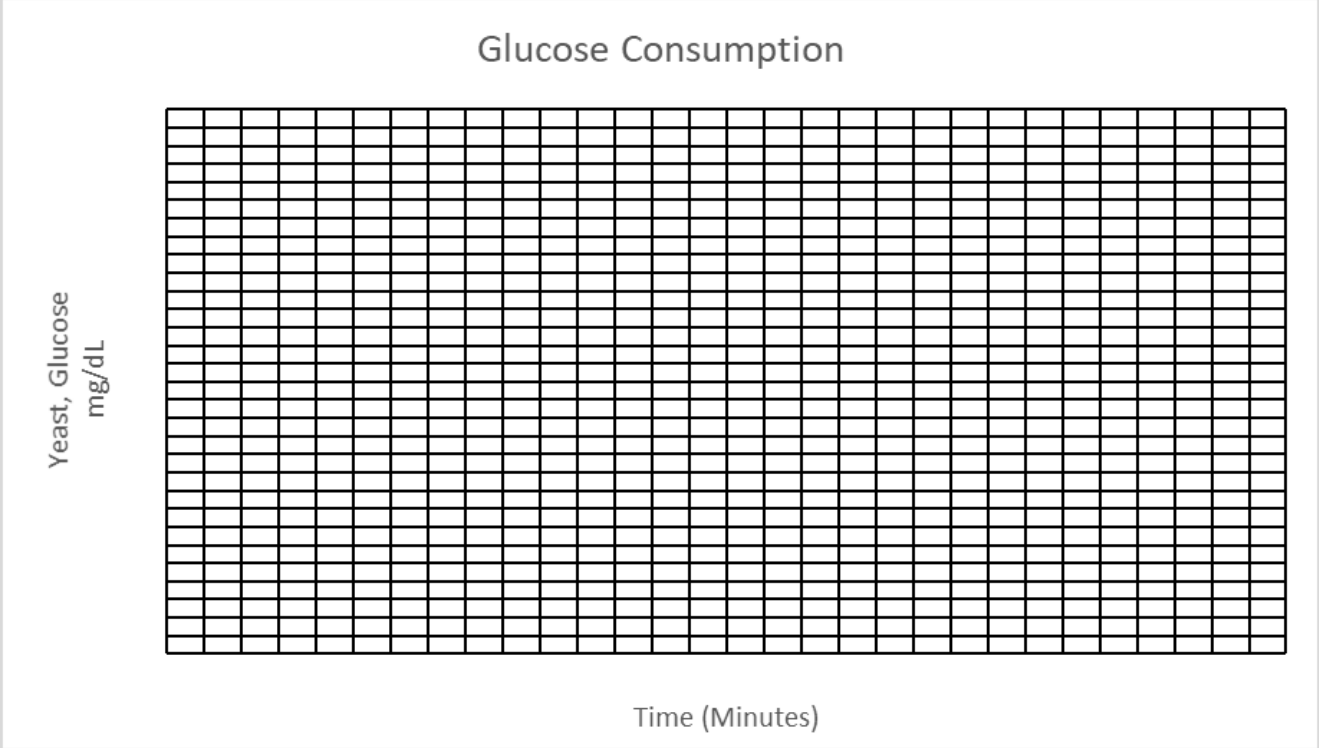
The meter starts counting down from 5, you can stop touching the drop.

9) Record the reading in the table below.
10) Dispose of the glucose strip.

11) Repeat testing glucose for control flask.
12) Record in the table your time of the readings and the glucose result. You will also record the glucose levels of fellow classmate.
13) Over 4 hours, you will test glucose in your flasks multiple times.

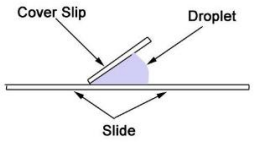
	Time:	Time:	Time:	Time:	Time:	Time:	Time:
0 grams							
0.5 grams							
1.0 grams							
1.5 grams							

Graph Results!

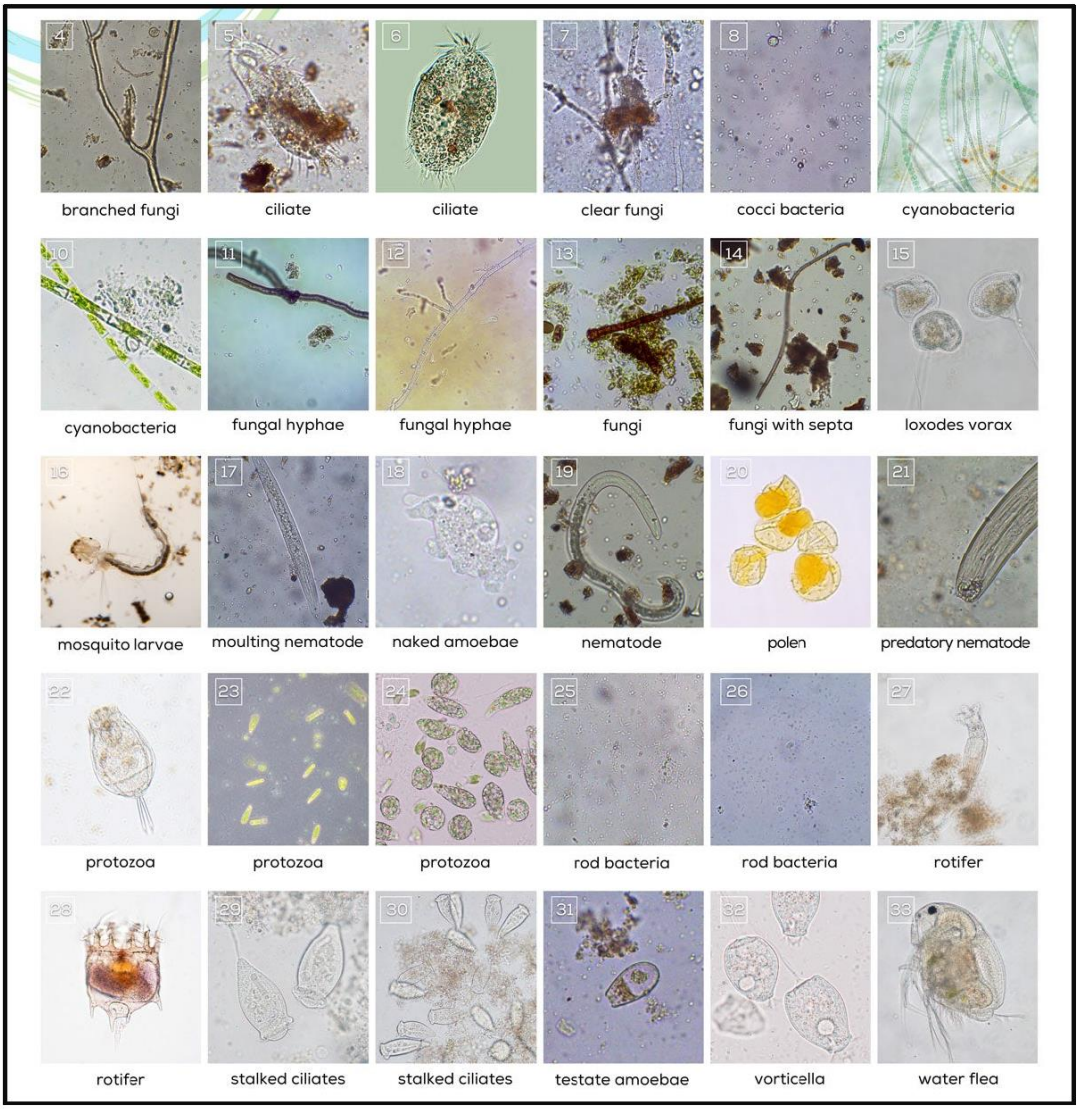


Secondary Biological Treatment Microscopy Lab

1. **Acquire gloves for safety.**
2. Using the sludge sample collected at the wastewater treatment plant, pipette 1 ml of sludge and place 1 drop onto the slide.
a. Be sure to mix your sludge sample well so it does not settle.
3. Using the slipcover, slide the cover along the side of the droplet and release.



a. Your droplet will spread between the lens and the slipcover.
4. Place the slide onto the microscope and compare the microorganisms to the following images below
5. Write down your results in your lab notebook.



Pondwater Microscopy Lab

1. **Acquire gloves for safety.**
2. Using the pond water sample, with a pipette place 2 drops onto the slide.
a. Be sure to mix your pond sample well so it does not settle.
3. Using the slipcover, slide the cover along the side of the droplet and release.
a. Your droplet will spread between the lens and the slipcover.
4. Place the slide onto the microscope and compare the microorganisms to the following images below
5. Write down your results in your lab notebook.

Colony Count

1) Take the petrifilm facing up.
2) Count the number of colonies in a square on the grid, and enter that number of colonies into the table below.
3) Repeat with four other squares on the grid, enter those values in the table below.
4) Determine the average bacterial count.

	Square 1	Square 2	Square 3	Square 4	Square 5	Sum	Average = Sum ÷ 5
Counts							

Gram Stain Instructions

Solutions:

1. Using the crystal violet staining reagent, place 2 drops, and wait **1 minute**.
2. Rinse the slide gently under a stream of tap water for 2 seconds.
3. Using the Gram's iodine mordant, pour the mordant for 3 seconds and **wait for 1 minute.**
4. Rinse the slide gently under a stream of tap water for 2 seconds.
5. Using the decolorizing agent, pour the agent for 3 seconds.
    a. **Wait 15 seconds** or add drop by drop to the slide until the decolorizing agent running from the slide becomes **clear**.
6. Using the counterstain (Safranin), pour the counterstain for 3 seconds and **wait 30 seconds to 1 minute.**
7. Rinse the slide gently under a stream of tap water until **no color** appears.
8. Dry the slide by blotting with a piece of absorbent paper.
9. Observe the results under oil immersion using a microscope.
    a. Pink/Red: gram-negative bacteria
    b. Blue/Purple: gram-positive bacteria

Water Filtration Engineering Challenge

Rapid Sand Filtration Test

Materials:

- Conical tubes (50ml)
- Sand (40 x 50)
- Small rock (60 x 80)
- Medium rock (80 x 120)
- Large rock (1/4 x 1/8)
- Funnel with PVC tubing attachment
- 1 liter of dirty sample water
- 1 liter plastic container (x2)
- Weigh boats
- Mass balance

Part 1: Experimenting

In part 1, you will be allowed to free test using the 50 ml conical tubes and the different rock media available. The conical tubes will be used to prepare different filter media combinations which could be used as your final design. The maximum FINAL mass of combined media needs to be less than or equal to 35 grams.

Use this time to test out different variations of media to create your unique filter. In addition, consider advantages and disadvantages of each design as well as cost and time to build and run your filters. **See cost analysis sheet to see the cost of your capital and operation and maintenance of your filter.** Be sure to test your different filters for initial and final turbidity, time, and odor (smell test).

Your objective is to reduce the turbidity to 1 NTU. You must have the final turbidity 1 NTU or less for 1 liter of sample water with lowest cost and maximum volume of filter media. Using your lab notebook, create a table for your different variations of filter media. Here is an example of a table:

Trials	Initial Turbidity	Final Turbidity	Time	Odor	Description: Media Order
1					
2					
3					
4					
5					
6					
7					
8					

Part 2: Constructing & Testing

1. At your filtering station, make sure your valve to your filter is closed.
2. Using your plastic container, add 50 ml of CLEAN water inside your column or fill until it reaches the line indicated on your column.
3. Begin adding your media to your column in the order your team has chosen. **NOTE: The media will need to be weighed using the mass balance. See mass balance instructions.**
4. After all your media is added, add an additional 200 ml of CLEAN water to your filter using your plastic container and place it under your filter. Allow the water to filter through until it reaches the line indicated on your column.
5. Close the valve once you start to see the water level in your column and pour the contents of your plastic container into the "waste" container. **DO NOT LET MEDIA DRY!**
6. Retrieve your 1-liter bottle of DIRTY water and shake well. Measure the **initial** turbidity and write your result.
    a. **See turbidimeter instructions. Record value in table below.**
7. Using your 1-liter bottle of DIRTY water, pour the entire contents into your filter and start your timer. Place plastic container under your filter.
8. Open the valve and allow to filter through.
9. Test for turbidity when your plastic container reaches 200ml. Record the Time.
10. Test for turbidity when your plastic container reaches 400 ml. Record the Time.
11. Test for turbidity when your plastic container reaches 600 ml. Record the Time.
12. Test for turbidity when your plastic container reaches 800ml. Record the Time.
13. With the remaining 200ml of filtered water in your plastic container, measure the **final** turbidity (at 1 L). Close the valve once you see the water level in the column. Record the Time.
    a. **See turbidimeter instructions. Record value in table below.**
14. In your lab notebook, write down your final time and turbidity results like the table below.

	Initial Turbidity (NTU) @ 200 ml	Turbidity (NTU) @ 400 ml	Turbidity (NTU) @ 600 ml	Turbidity (NTU) @ 800 ml	Final Turbidity (NTU) @ 1 L
Column Test					
Time (min)					

Part 3: Testing Analysis

1. Prepare a table showing your cost analysis and filter run time for your filter. *See cost analysis page 5.*
2. Create a bar graph to show turbidity performances for **each** of your trials from part 1 of experimentation.
    a. **NOTE: Be sure to have all trials on the same graph**
3. Create another bar graph showing the time of **each** trial filter prepared in part 1 of experimentation.
    a. **NOTE: Be sure to have all trials on the same graph**
4. Prepare a bar graph showing your actual design turbidity results from your filter in part 2 of the laboratory.
    a. **NOTE: Be sure to have initial turbidity and final turbidity**

Part 4: Presentation

Prepare a 3 – minute presentation that includes the following:

- Selected media (mass and grain size)
- Media order
- Total cost
- Prepared tables and graphs
- Final effluent turbidity

In your presentation, be sure to discuss your thought process. What were some advantages and disadvantages you noticed? How could you improve your filter?

Be creative! Come up with a team name! Add pictures, videos, graphics, clip art, animations, and transitions. Also, be sure to label all tables and graphs.

Cost Analysis

Subject	*Conditions*	*Equations*	*Variables*
Filter Run Time Cost (O&M)	$0 < \mathbf{A} < 30$: $\mathbf{A} \geq 30$:	$\mathbf{Z_1} = \$10.00$ $\mathbf{Z_1} = \mathbf{A}\text{-}20$	**A = time** **Z_1 = cost**
Capital Cost for Each Media	Sand (40 x 50) Small rock (60 x 80) Medium Rock (80 x 120) Large Rock (80 x 120)	$\mathbf{Z_2} = \$2.00$/gram $\mathbf{Z_2} = \$1.75$/gram $\mathbf{Z_2} = \$1.25$/gram $\mathbf{Z_2} = \$1.00$/gram	**ΣZ_2 = cost**
Turbidity Penalty Cost	$\mathbf{T} < 1$: $\mathbf{T} > 50$: $1 < \mathbf{T} < 50$:	$\mathbf{Z_3} = \$0.00$ $\mathbf{Z_3} = \$100.00$ $\mathbf{Z_3} = 1.75\mathbf{T}$	**T = Turbidity** **Z_3 = Cost**

	Final Costs
Filter Run Time Cost **(Z_1)**	$
Capital Cost for Each Media **(Sum of Z_2 = ΣZ_2)**	$
Turbidity Penalty Cost **(Z_3)**	$
SUM ($Z_1 + \Sigma Z_2 + Z_3$)	$

My College and Career Preparation Plan *Part 1 Careers*

Name _____ Age _____ Grade _____

Resources: https://bls.gov/k12/students/ and https://myfuture.com/career

Careers of Interest	Required Training/Schooling	Core Classes Needed	Skills Emphasized	What will I do in this Career?

HWCA: My College and Career Preparation Plan *Part 2 High School Timeline for College/Career Preparation*

Resources: https://myfuture.com/college/college-planning-timeline and https://act.org/content/act/en/students-and-parents/high-school-success/high-school-resources.html

Things to Consider and Do	Freshman	Sophomore	Junior	Senior
High School Classes to Take				
Testing ACT SAT/PSAT				
FAFSA Financial Aid				
Extracurricular Activities and Accomplishments				
College Application Needs				
Other Stuff				

Schools of Interest _____ _____ _____ _____

Baltic WWTP Glossary
https://www.balticwaterhub.net/tool/WWTP-game

Key Terms	Definitions
Aerobic	uses oxygen to break down organic contaminants and other pollutants like nitrogen (N) and phosphorous (P).
Anaerobic	breaks down organic contaminants and pollutants without oxygen.
Activated Sludge	a large group of microorganisms used to break down organic matter in an aeration basin (with oxygen).
BOD7 (Biological Oxygen Demand)	a measure of the amount of oxygen needed for microorganisms to break down the organic matter within seven days.
Flocculation	slow-mixing which allows for larger particles (floc-formation) to stick together, become heavier, and settle out by gravity.
F/M	a ratio used to show the amount of food entering the system versus the number of microorganisms in the system.
Lime $Ca(OH)_2$	Used in wastewater to raise the pH to help settle out metal salts into solids to be removed.
Metal Salts	used as a coagulant or chemical substance to help balance (neutralize) the positive and negative ions in the system.
MLSS (Mixed Liquor Suspended Solids)	the amount of active biomass available to consume the organics.
Organic Matter	matter that has come from living organisms.
Precipitants	the change from dissolved material into solid particles
SVI (Sludge Volume Index)	settleability of sludge.